

JavaScript Errors

JavaScript errors indicate that something went wrong during parsing or execution.

Being able to identify the error type quickly is essential for exams, interviews, and real-world debugging.

Core concepts:

- Errors are **objects**
- Every error has a **type**, a **message**, and a **stack trace**
- Errors can happen at **parse time** or **runtime**

```
Error
error.name
error.message
error.stack
```

Node traversal (includes text & comments)

SyntaxError

- **Meaning:** The JavaScript engine cannot parse the code
- **Occurs when:** Invalid syntax, malformed JSON
- **Example:** `JSON.parse("{a:1}")`

ReferenceError

- **Meaning:** A variable or identifier does not exist
- **Occurs when:** Using undeclared or out-of-scope variables
- **Example:** `console.log(foo)`

TypeError

- **Meaning:** Operation is not valid for the given type
- **Occurs when:** Calling non-functions, accessing properties on
- **Occurs when:** Calling non-functions, accessing properties on `null` or `undefined`
- **Example:** `null.name`

RangeError

- **Meaning:** A value is outside the allowed range
- **Occurs when:** Invalid array length, recursion limit exceeded
- **Example:** `new Array(-1)`

URIError

- **Meaning:** Malformed URI sequence
- **Occurs when:** Invalid `encodeURIComponent` / `decodeURIComponent` usage
- **Example:** `decodeURIComponent('%')`



EvalError

- **Meaning:** Error related to `eval()`
- **Occurs when:** Legacy / deprecated eval usage (rare)
- **Example:** `eval("")`

AggregateError

- **Meaning:** Multiple errors grouped together
- **Occurs when:** `Promise.any()` fails with multiple rejections
- **Example:** `Promise.any([Promise.reject(), Promise.reject()])`

InternalError (engine-specific)

- **Meaning:** JavaScript engine internal failure
- **Occurs when:** Stack overflow, extreme recursion
- **Example:** `function f(){ f() } f()`

DOMException (browser only)

- **Meaning:** Error thrown by Web APIs
- **Occurs when:** Invalid DOM operations, security or permission issues
- **Example:** `navigator.clipboard.readText()`

Error object properties

`name` → error type (`TypeError` , `ReferenceError` , ...)

`message` → human-readable explanation

`stack` → execution trace

```
error.name
error.message
error.stack
```

Ultra-short mental model

- Errors = **signals**
- Type tells you **why**
- Message tells you **what**
- Stack tells you **where**

One-line memory map

`SyntaxError` → invalid syntax

`ReferenceError` → unknown variable

`TypeError` → invalid type usage

`RangeError` → out of bounds

`URIError` → malformed URI

`EvalError` → eval-related (legacy)

`AggregateError` → multiple async failures

`InternalError` → engine limit

`DOMException` → browser API error